Equilibrium Quiz 8 **Time: 15 mins** **Mark: ____/12** **_____%**

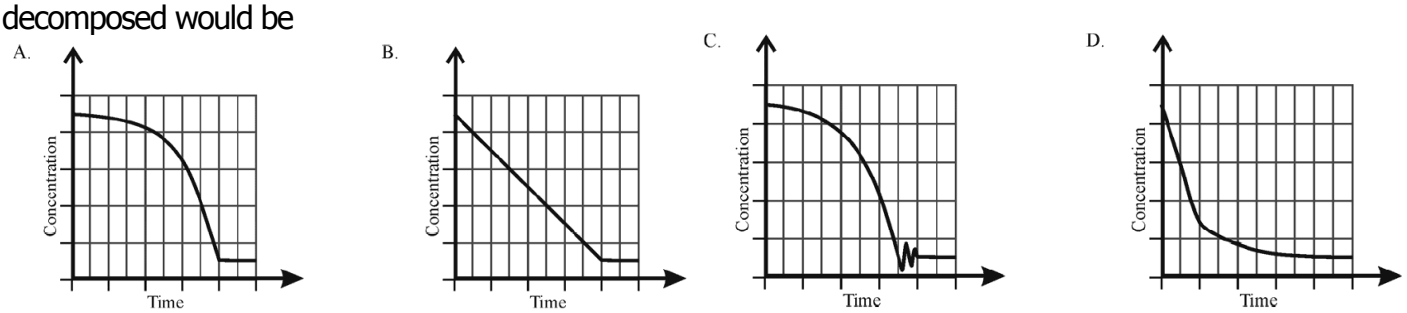
Question 1
In an endothermic reaction

a) the activation energy of the forward reaction is less than the activation energy of the reverse reaction.
b) the energy released when the bonds in the products are formed is greater than that required to break the bonds in the reactants.
c) the rate of the reverse reaction will increase as the temperature is lowered.
d) the energy required to break the bonds in the reactants is greater than that released when the bonds in the products are formed.

Question 2
The decomposition of nitrogen (V) oxide can be represented by the chemical equation

$$N_2O_{5(g)} \rightleftharpoons 2NO_{2(g)} + \tfrac{1}{2}O_{2(g)}$$

The graph that would best represent how the concentration of nitrogen (V) oxide varies with time as it decomposed would be



Question 3
The expression for the equilibrium constant for a chemical reaction is $K = \frac{[NOCl]^2}{[NO]^2[Cl_2]}$

The chemical equation that would represent this reaction would be

a) $2NO_{(g)} + Cl_{2(g)} \rightleftharpoons 2NOCl_{(g)}$
b) $NOCl_{(g)} \rightleftharpoons NO_{(g)} + \tfrac{1}{2}Cl_{2(g)}$
c) $2NOCl_{(g)} \rightleftharpoons 2NO_{(g)} + Cl_{2(g)}$
d) $NO_{(g)} + \tfrac{1}{2}Cl_{2(g)} \rightleftharpoons NOCl_{(g)}$

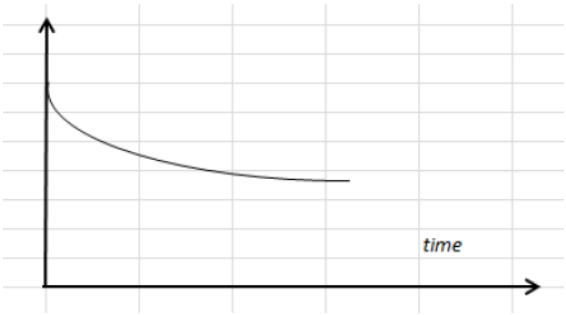
Question 4
One source of hydrogen is from methane.

$$CH_{4(g)} + H_2O_{(g)} \rightleftharpoons CO_{(g)} + 3H_{2(g)} \quad \Delta H = -206 \text{ kJ mol}^{-1}$$

The greatest increase in the yield of this reaction would be from

a) the addition of a catalyst and an increase in pressure
b) an increase in temperature and a decrease in pressure
c) an increase in temperature and an increase in pressure
d) a decrease in temperature and a decrease in pressure

Question 5
The following graph is drawn from data collected from a reaction of marble chips, $CaCO_3$, reacting in a flask with hydrochloric acid.



The graph could represent the

a) mass of the gas evolved from the flask
b) pH of the flask contents
c) volume of the gas evolved from the flask
d) mass of the flask over a period of time

Question 6
When the ion OCl^- is in an aqueous solution it will undergo hydrolysis according to the equation:

$$OCl^-_{(aq)} + H_2O_{(l)} \rightleftharpoons HOCl_{(aq)} + OH^-_{(aq)}$$

A few drops of acid are added to an OCl^- solution. The addition of the hydrochloric acid will

a) push the reaction in the forward direction and the pH will now be lower
b) push the reaction in the forward direction and the pH will now be higher
c) have no effect on the reaction as the equation does not contain H_3O^+ ions
d) push the reaction in the reverse direction as the acid reacts with the water

Question 7
The reaction for sulfuric acid in water occurs in two stages;

$$H_2SO_{4(aq)} + H_2O_{(l)} \rightarrow H_3O^+_{(aq)} + HSO_4^-{}_{(aq)}$$

$$HSO_4^-{}_{(aq)} + H_2O_{(l)} \rightleftharpoons H_3O^+_{(aq)} + SO_4^{2-}{}_{(aq)} \qquad K_a = 1.2 \times 10^{-2} M \text{ at } 25^oC$$

From this information we can conclude that the pH of a 0.10M solution of sulfuric acid will be

a) 0
b) 0.7
c) Between 0.7 and 1.0
d) 1.0

Question 8
An equilibrium reaction is represented by the chemical equation $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)}$ $\Delta H > 0$

An equimolar mixture of the reactants is placed in a piston and allowed to come to equilibrium. The volume of the piston is decreased to one quarter of its original volume and allowed to re-establish equilibrium while keeping the temperature constant. Compared to the initial equilibrium mixture the amount, n(B), and concentration, [B], under the new conditions will be

	n(B)	**[B]**
a)	Higher	Lower
b)	Lower	Higher
c)	Higher	Higher
d)	Lower	Lower

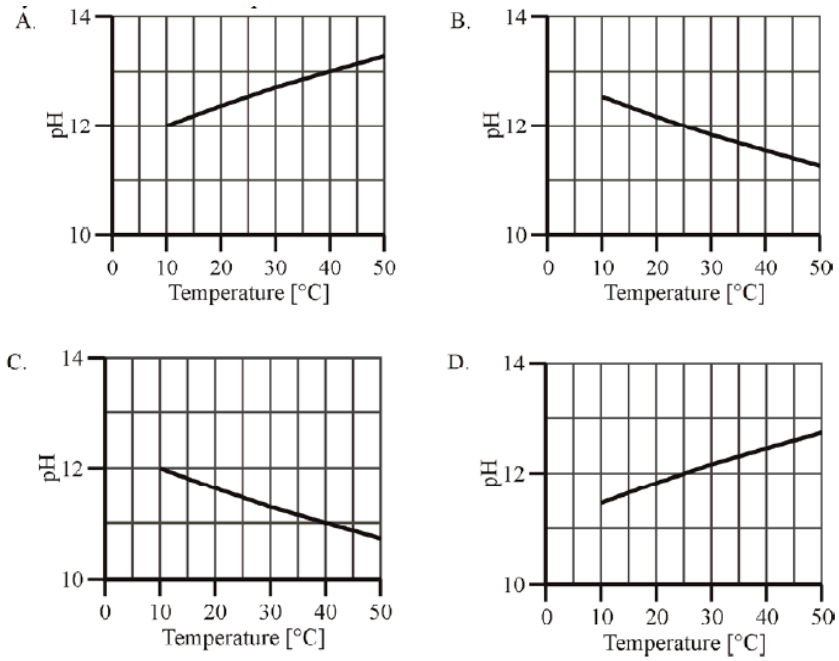
Question 9
An exothermic chemical reaction involved in an industrial process is carried out in the presence of a catalyst and at elevated temperature and pressure. The conditions used in the process suggest that

a) The value for the equilibrium constant at SLC is lower than at the higher temperature
b) The rate of the forward reaction in the absence of a catalyst is slow
c) When the reactants are mixed together they will readily reach equilibrium
d) The rate of the forward reaction in the absence of the catalyst is fast

Question 10
The self-ionisation of water can be described by the chemical equation

$$2H_2O_{(l)} \rightleftharpoons H_3O^+_{(aq)} + OH^-_{(aq)} \qquad \Delta H = 56\ kJmol^{-1}$$

The graph that would best describe how the pH of a 0.010M aqueous solution of sodium hydroxide varies with temperature would be



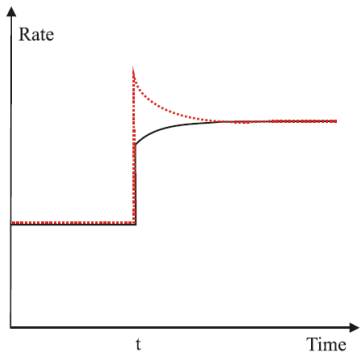
Question 11
Enthalpy changes for two reactions of copper are shown below;

$$Cu_{(s)} \rightarrow Cu^+_{(aq)} \qquad \Delta H = +602kJmol^{-1}$$
$$Cu_{(s)} \rightarrow Cu^{2+}_{(aq)} \qquad \Delta H = +795kJmol^{-1}$$

Use these values to calculate the enthalpy change for the reaction: $2Cu^+_{(aq)} \rightarrow Cu_{(s)} + Cu^{2+}_{(aq)}$

a) $+193kJmol^{-1}$
b) $-193kJmol^{-1}$
c) $-409kJmol^{-1}$
d) $+409kJmol^{-1}$

Question 12
The impact of a change imposed, at time **t**, on an $2NO_{2(g)} \rightleftharpoons N_2O_{4(g)}$ equilibrium mixture is shown on the rate-time graph below.



Dotted line = forward reaction

Solid line = reverse reaction

The change imposed on the system was the

a) Addition of $NO_{2(g)}$.
b) Addition of $N_2O_{4(g)}$.
c) Increase of container volume.
d) Decrease of container volume.

Answer Key

Question 1
In an endothermic reaction

a) the activation energy of the forward reaction is less than the activation energy of the reverse reaction.
b) the energy released when the bonds in the products are formed is greater than that required to break the bonds in the reactants.
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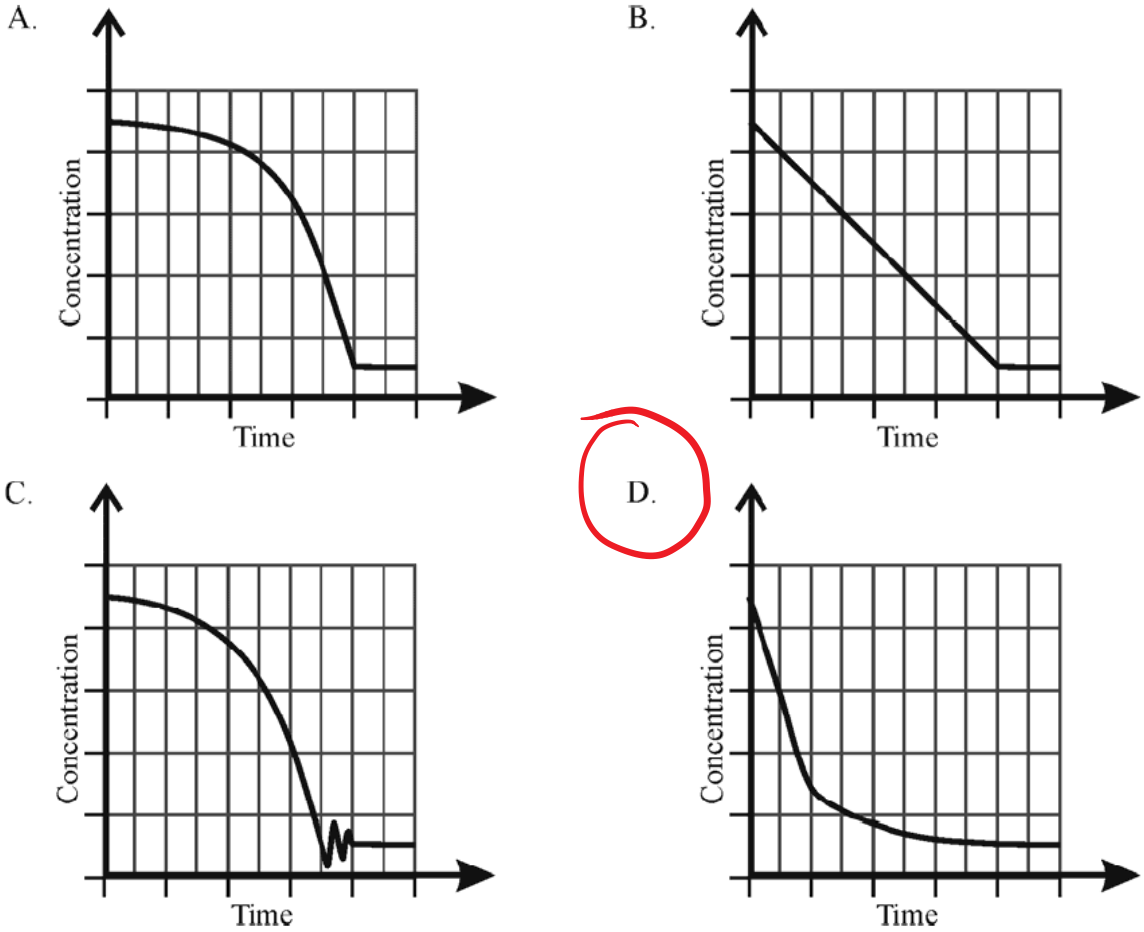

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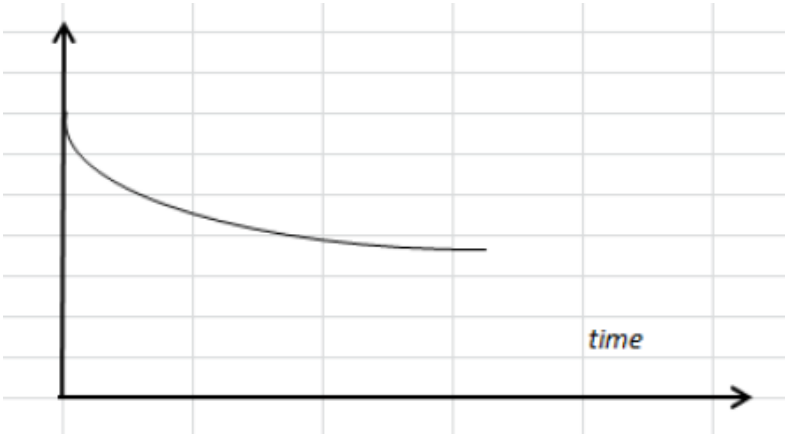

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	n(B)	[B]
A	Higher	Lower
B	**Lower**	**Higher**
C	Higher	Higher
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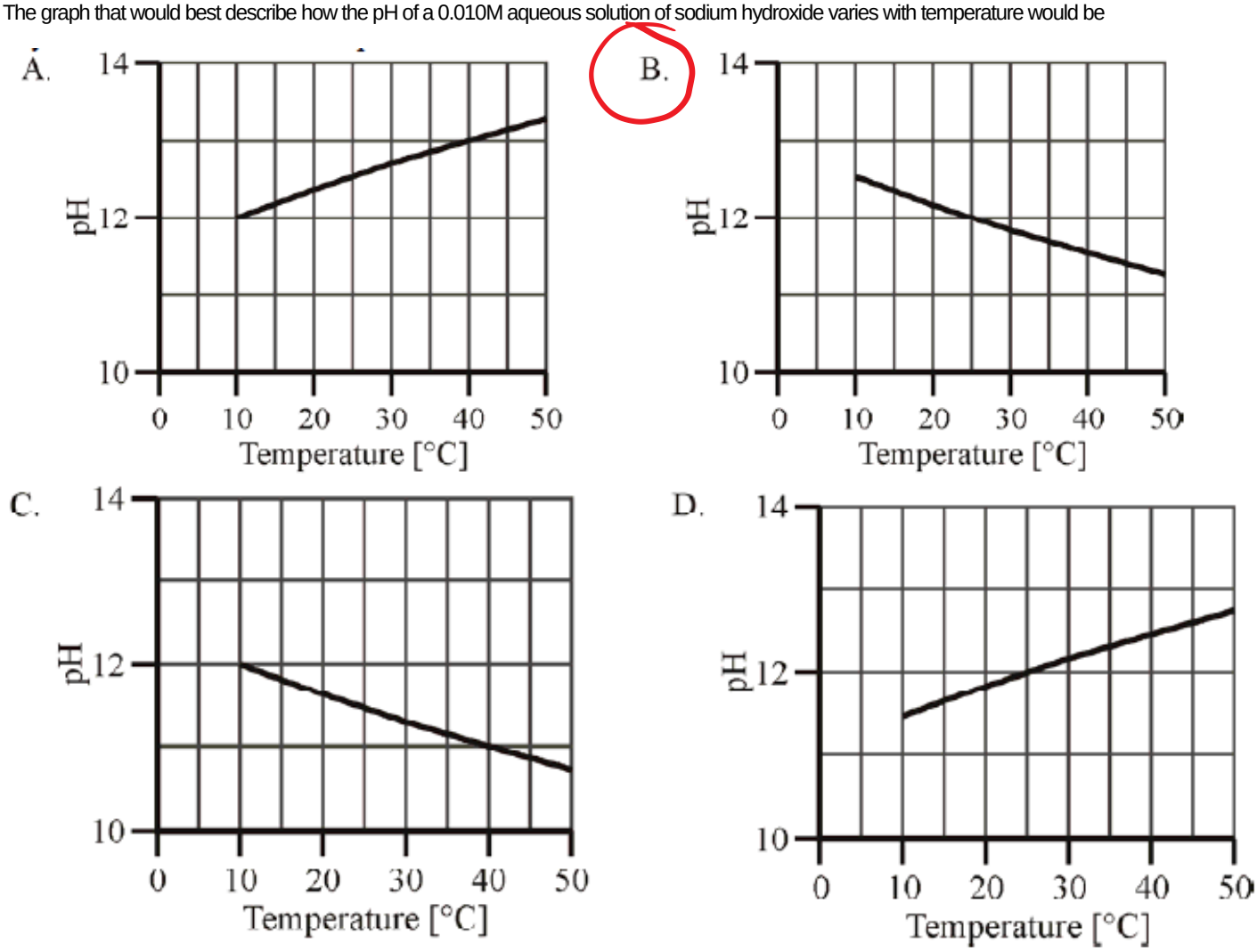

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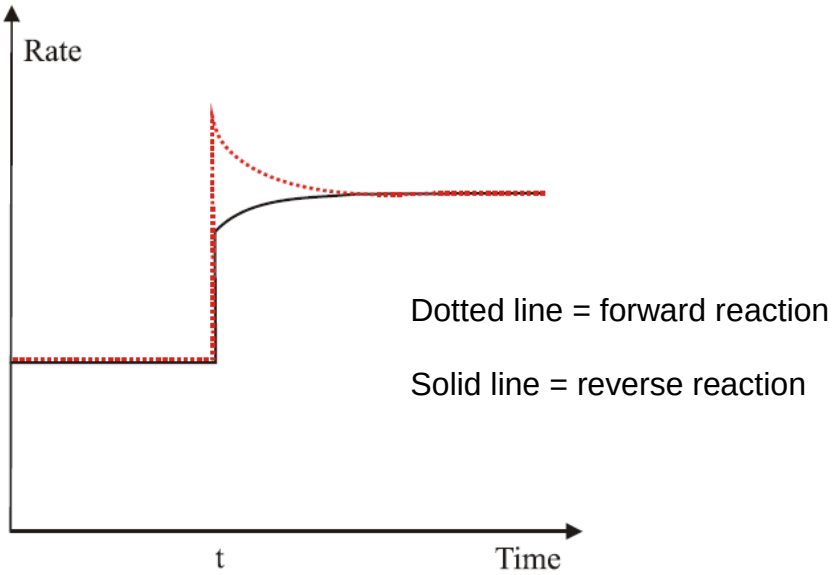

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12 marks